

No. of Functional Features Included in the Solution

Date	7 July, 2026
Team ID	
Project Name	Machine Learning-Based Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Functional Features Overview

This document lists all the functional features implemented in the Credit Card Approval Prediction project. Each feature contributes to automating the credit card approval process using Machine Learning and a Flask web application.

Functional Features

S.No	Feature Name	Feature Description	Module / Component	Status (Done / in progress / Pending)	Marks Contribution
1	Weather Data Input	Accepts weather parameters such as annual rainfall, seasonal rainfall, and cloud visibility through a web form..	Flak UI	Done	High
2	Data Preprocessing	Cleans, validates, and preprocesses weather data before prediction..	Data processing	Done	High
3	Flood Prediction	Predicts whether flood occurrence is Likely or Not Likely using the trained ML model.	Prediction Engine	Done	Medium
4	Model Integration	Loads the trained XGBoost model into the Flask application using Pickle.	Backend	Done	Medium
5	User-Friendly Web Interface	Displays flood prediction results instantly through an interactive web application.	Frontend	Done	Medium

Feature Summary:

Metric	Count / value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Displays flood prediction results instantly through an interactive web application.
Additional / Nice-to-have Features	User-Friendly Interface, Model Evaluation, Error Handling, Model Serialization
Features Tested & Verified	8 Features Successfully Tested

Feature Category Breakdown

S.No	Category	Features in Category	Example Feature
1	User Interface(UI)	2	Weather Data Input Form, Flood Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, Prediction Engine, Input Validation
3	Database / Storage	1	Weather Dataset Management and Pickle Model Storage
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	1	Input Validation and Exception Handling